

Mock Lab Test — Round 1

Rules:

- **Q1–Q10:** Select **ONE** answer.
- **Q11–Q20:** Select **ALL correct answers**. There may be **1, 2, 3, or more** correct answers.
- Don't worry if you're unsure—answer everything first, then I'll mark and explain each one.

Q1 — Train/Test Split

A dataset contains **500 samples**. You use:

```
train_test_split(X, y, test_size=0.2)
```

How many samples will be in the test set?

- A. 50
- B. 100
- C. 200
- D. 400

Q2 — Data Leakage

Person P does this:

1. Scale the entire dataset
2. Split into training and testing sets

Person Q does this:

1. Split the dataset
2. Fit the scaler on training data
3. Transform training and testing data using that scaler

Which statement is correct?

- A. Person P is correct because more data improves scaling.
- B. Person Q is correct because Person P causes data leakage.
- C. Both methods are correct.
- D. Neither method is correct.

Q3 — Scaling

Which scaler typically transforms features into the range **0 to 1**?

- A. StandardScaler
- B. MinMaxScaler
- C. LabelEncoder
- D. OneHotEncoder

Q4 — StandardScaler

Which statement best describes StandardScaler?

- A. It transforms values to the range 0 to 1.
- B. It removes duplicate rows.
- C. It transforms data to approximately mean 0 and standard deviation 1.
- D. It converts categorical values into numbers.

Q5 — Encoding

A feature called Color contains:

Red

Blue

Green

These categories have **no natural order**.

Which encoding method is generally appropriate?

- A. One-Hot Encoding
- B. StandardScaler
- C. MinMaxScaler
- D. Log Transformation

Q6 — Scatter Plot and Correlation

In a scatter plot, as X increases, Y generally increases, but the points do not lie exactly on a straight line.

What does this most likely indicate?

- A. Perfect positive correlation
- B. Positive correlation, but not perfect
- C. Negative correlation
- D. No correlation

Q7 — Outliers

Which plot is generally best for quickly identifying **outliers in a numerical variable**?

- A. Pie chart
 - B. Box plot
 - C. Line chart
 - D. Bar chart
-

Q8 — PCA and Scaling

You have two features:

Feature 1: values between 0 and 10

Feature 2: values between 0 and 100,000

Before applying PCA, what is generally the best approach?

- A. Remove Feature 2
 - B. Scale/standardize the features
 - C. Convert both features to categorical variables
 - D. Duplicate Feature 1
-

Q9 — Logistic Regression

Which equation represents the **linear combination before applying the sigmoid function**?

A.

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

B.

$$z = Xw + b$$

C.

$$d = \sqrt{\sum (x_i - y_i)^2}$$

D.

$$\theta = \theta - \alpha \nabla J(\theta)$$

Q10 — Euclidean Distance

Calculate the Euclidean distance between:

$$P = (1, 2)$$

and

$$Q = (4, 6)$$

- A. 3
 - B. 4
 - C. 5
 - D. 7
-

Checkbox Questions — Select ALL Correct Answers

Q11 — Data Balancing

Which techniques can be used to deal with an imbalanced dataset?

- ☐ A. Random oversampling
- ☐ B. Random undersampling
- ☐ C. SMOTE
- ☐ D. StandardScaler

Select all correct answers.

Q12 — Overfitting / Underfitting

Which statements are correct?

- ☐ A. An underfitted model may perform poorly on both training and testing data.
- ☐ B. An overfitted model usually performs very well on training data but poorly on unseen data.
- ☐ C. A good-fit model always has 100% training accuracy.
- ☐ D. Underfitting can indicate that the model is too simple.

Select all correct answers.

Q13 — Stratified Split

Which statements about a **stratified train-test split** are correct?

- ☐ A. It attempts to preserve class proportions in training and testing sets.
- ☐ B. It is especially useful for classification with imbalanced classes.
- ☐ C. It guarantees that every feature has exactly the same distribution.
- ☐ D. It can be specified using the stratify parameter in `train_test_split()`.

Select all correct answers.

Q14 — PCA

Which statements about PCA are correct?

- ☐ A. PCA is commonly used for dimensionality reduction.
- ☐ B. Principal components are selected based on the amount of variance they explain.
- ☐ C. PCA always increases the number of features.
- ☐ D. Features with very different scales may cause problems if PCA is applied without scaling.

Select all correct answers.

Q15 — Gradient Descent

Consider:

$$\theta := \theta - \alpha \frac{\partial J}{\partial \theta}$$

Which statements are correct?

- ☐ A. α is the learning rate.
- ☐ B. The gradient indicates the direction of increase of the cost function.
- ☐ C. Gradient descent subtracts the gradient to move toward a minimum.
- ☐ D. A very large learning rate may cause overshooting or divergence.

Select all correct answers.

Q16 — kNN

Which statements about kNN are correct?

- ☐ A. kNN makes predictions based on nearby training examples.
- ☐ B. Feature scaling is important because kNN uses distances.
- ☐ C. A very small K can be more sensitive to noise.
- ☐ D. kNN uses the sigmoid function to classify data.

Select all correct answers.

Q17 — Odd Values of K

Why is an odd value of K often used in binary classification?

- ☐ A. It can help reduce the chance of a voting tie.
- ☐ B. Odd K always gives better accuracy than even K.
- ☐ C. With $K = 3$, a binary vote cannot result in a 1.5 vs 1.5 tie.
- ☐ D. Odd K guarantees that overfitting will never happen.

Select all correct answers.

Q18 — Logistic Regression and Log Loss

Which statements are correct?

- ☐ A. Logistic Regression commonly uses the sigmoid function to produce values between 0 and 1.
- ☐ B. Log Loss/Binary Cross-Entropy is commonly used for binary classification.
- ☐ C. Logistic Regression predicts classes by calculating Euclidean distance to the nearest point.
- ☐ D. A threshold such as 0.5 can be used to convert a probability into a class prediction.

Select all correct answers.

Q19 — Duplicates

Suppose a dataset contains duplicate rows.

Which statements are correct?

- ☐ A. `df.duplicated()` can help identify duplicate rows.
- ☐ B. `df.drop_duplicates()` can remove duplicate rows.
- ☐ C. Duplicate rows can sometimes bias a model if they unnecessarily repeat observations.
- ☐ D. `StandardScaler` automatically removes duplicate rows.

Select all correct answers.

Q20 — SMOTE vs Scaling

Which statements are correct?

- ☐ A. SMOTE is a technique used to address class imbalance.
- ☐ B. `StandardScaler` is primarily used to balance class distributions.
- ☐ C. SMOTE creates synthetic minority-class samples.
- ☐ D. Scaling and data balancing solve exactly the same problem.

Select all correct answers.